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AI HYBRID BASED PLAGIARISM DETECTION SYSTEM CREATION

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INTRODUCTION

- The rapid growth of digital information and easy access to online resources have increased the risk of plagiarism in academia.
- Plagiarism, whether intentional or unintentional, threatens academic integrity by misusing others' ideas without proper acknowledgment. Traditional plagiarism detection methods—such as manual checking and rule-based systems—are inefficient and lack accuracy when dealing with large-scale and complex data.
- To address this issue, Artificial Intelligence (AI), particularly Machine Learning (ML) and Natural Language Processing (NLP), is introduced to enhance plagiarism detection with better accuracy, speed, and scalability.

OBJECTIVES

- To design and implement an intelligent system using machine learning and NLP techniques. The system aims to automatically analyze large volumes of text data. It focuses on identifying similarities between documents efficiently.
- To overcome limitations of traditional rule-based and manual detection methods. The system ensures faster processing with higher precision and recall. It enhances scalability for handling large academic datasets.
- To identify various forms such as direct copying, paraphrasing, and citation manipulation. The system analyzes both surface-level and deep semantic similarities. It ensures comprehensive detection across different writing styles.

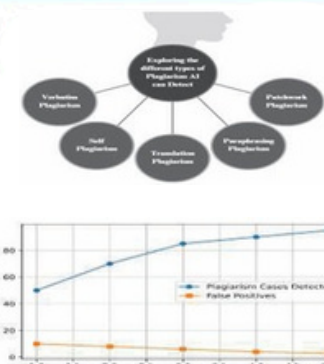
METHODOLOGY

- **Data Collection & Preprocessing:** Collect large datasets of academic documents from reliable sources. Clean the text using tokenization, stop-word removal, and stemming. Prepare structured data for further analysis.
- **Feature Extraction:** Extract important features like word frequency (lexical), sentence structure (syntactic), and meaning (semantic). Use techniques such as TF-IDF and word embeddings. Convert text into numerical format for machine learning models.
- **Model Training:** Train supervised ML models like SVM, Random Forest, and Gradient Boosting. Use labeled datasets to distinguish between original and plagiarized content. Optimize model performance using validation techniques.
- **Detection & Reporting:** Highlight plagiarized sections and calculate similarity percentage. Generate detailed reports with sources and overlap information. Provide real-time feedback for users.
- **Deployment & Improvement:** Integrate the system with LMS or web applications. Continuously update the model using user feedback. Ensure ethical practices like data privacy and bias reduction.

ARCHITECTURE

The system architecture consists of the following components:

- Input Layer – User uploads document
 - Preprocessing Module – Text cleaning and normalization
 - Feature Extraction Module – Converts text into numerical representations
 - Similarity Analysis Module – Computes similarity scores
 - Machine Learning Model – Classifies plagiarism
 - Database/Corpus – Stores reference documents
 - Output Layer – Generates plagiarism report with highlighted sections
- This architecture ensures scalability, efficiency, and accurate detection.



RESULT AND CONCLUSION

- AI-powered plagiarism detection systems significantly improve the reliability and efficiency of detecting academic misconduct. By leveraging ML and NLP techniques, these systems can detect even subtle forms of plagiarism.
- However, challenges such as false positives, ethical concerns, and algorithmic bias must be carefully managed. Overall, AI serves as a powerful tool to promote originality, transparency, and academic integrity when combined with proper ethical guidelines and human oversight.

REFERENCES

- [1] Pecorari, D. (2018). Plagiarism. In Handbook of English for Specific Purposes)
- [2] Roig, M. (2019). Plagiarism and self-plagiarism: What every author should know. Biochemia Medica, '
- [3] Stamatatos, E. (2009). A survey of modern authorship attribution methods. Journal of the American Society for Information Science and Technology,